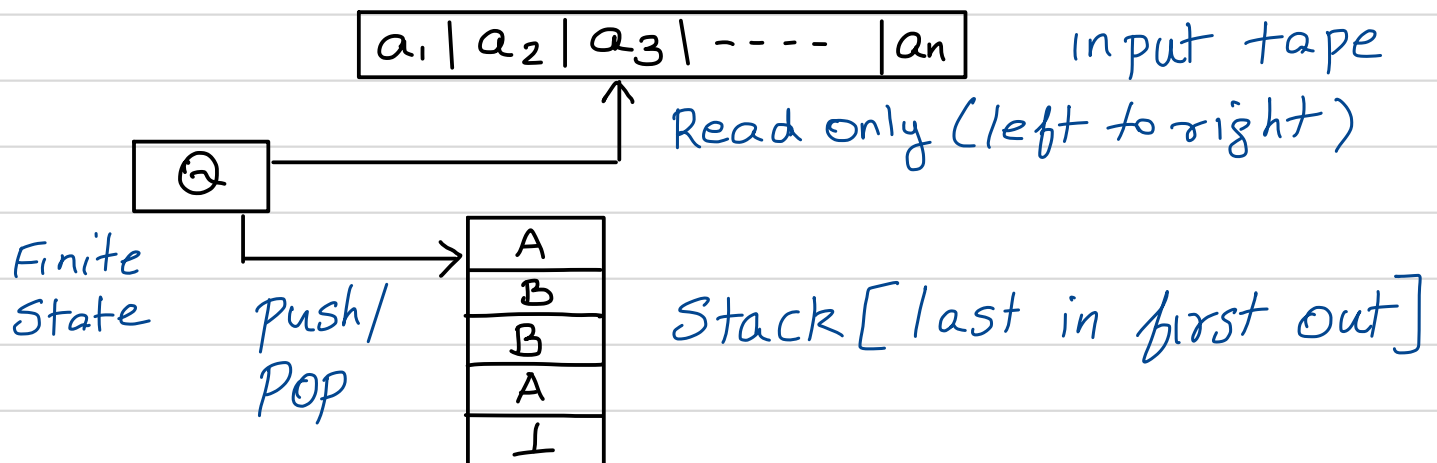


Non deterministic pushdown automata (NPDA)

Finite automata + a stack.



Working of the machine

- pops the top symbol off the stack.
- Makes a transition based on the top of the stack, input symbol and current state.

Transition: push a sequence of symbols onto the stack, change state, move the read head one cell to the right

ϵ -transitions are allowed: Machine can pop and push without reading an input symbol or moving the input head pointer.

Stack can store unbounded information but access is limited

Acceptance. Two types: By final state and empty stack.

By final state: M accepts x by final state if

$$(q, x, \perp) \xrightarrow{*}_M (q, \epsilon, \gamma) \text{ for some } q \in F, \gamma \in \Gamma^*$$

$\hookrightarrow$ can be any string.

By empty stack. M accepts x by empty stack if

$$(q, x, \perp) \xrightarrow{*}_M (q, \epsilon, \epsilon) \text{ for some } q \in Q.$$

$\hookrightarrow$ can be any state

$L(M)$ - set of all strings $x \in \Sigma^*$ accepted by M .

$$L = \{ w w^R \mid w \in \{a, b\}^* \}$$

$$M = (\{q_0, q_1, q_2\}, \{a, b\}, \{a, b, \perp\}, \delta, q_0, \perp, F)$$

$F = \{q_2\}$ - Accepting by final state.

$$- ((q_0, a, \perp), (q_0, a\perp)) \in \delta ; ((q_0, b, \perp), (q_0, b\perp)) \in \delta$$

$$- ((q_0, a, a), (q_0, aa)) ; ((q_0, a, b), (q_0, ab)) \left. \vphantom{\begin{matrix} ((q_0, a, a), (q_0, aa)) \\ ((q_0, a, b), (q_0, ab)) \end{matrix}} \right\} \in \delta$$

$$((q_0, b, a), (q_0, ba)) ; ((q_0, b, b), (q_0, bb)) \left. \vphantom{\begin{matrix} ((q_0, b, a), (q_0, ba)) \\ ((q_0, b, b), (q_0, bb)) \end{matrix}} \right\} \in \delta$$

$$- ((q_0, \epsilon, a), (q_1, a)) ; ((q_0, \epsilon, b), (q_1, b)) \left. \vphantom{\begin{matrix} ((q_0, \epsilon, a), (q_1, a)) \\ ((q_0, \epsilon, b), (q_1, b)) \end{matrix}} \right\} \in \delta$$

$$((q_0, \epsilon, \perp), (q_1, \perp)) \left. \vphantom{((q_0, \epsilon, \perp), (q_1, \perp))} \right\} \in \delta$$

$$- ((q_1, a, a), (q_1, \epsilon)) ; ((q_1, b, b), (q_1, \epsilon))$$

$$- ((q_1, \epsilon, \perp), (q_2, \epsilon))$$

NPDA's and CFGs have equivalent expressive power

Theorem 1. Given a CFG G , we can construct an NPDA M s.t. $L(G) = L(M)$.

Theorem 2. Given an NPDA M , we can construct a CFG G s.t. $L(G) = L(M)$.

Intersection of a CFL and a regular set

Theorem. CFLs are closed under intersection with regular sets.

if $A \subseteq \Sigma^*$ is a CFL and $B \subseteq \Sigma^*$ is a regular set
then $A \cap B$ is a CFL.

Proof idea. Consider an NPDA M_1 and a DFA M_2 s.t.
 $L(M_1) = A$ and $L(M_2) = B$

NPDA N : Apply the product construction on M_1 and M_2

- States of N are product of states of M_1 and M_2
- Stack of N simulates stack of M_1

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An application

$$A = \{w \in \{a, b, c\}^* \mid \#_a(w) = \#_b(w) = \#_c(w)\}$$

$$B = A \cap L(a^*b^*c^*) = \{a^n b^n c^n \mid n \geq 0\}$$

if A is a CFL then by Theorem B is a CFL
which is a contradiction.

Deterministic PDA.

$$M = (Q, \Sigma, \Gamma, \delta, \perp, \textcolor{blue}{-|}, \delta, F)$$

- $-|$ is a special symbol not in Σ . Right end marker

$$\delta \subseteq (Q \times (\Sigma \cup \{-|\} \cup \{\epsilon\}) \times \Gamma) \times (Q \times \Gamma^*)$$

- For any $p \in Q$, $a \in \Sigma \cup \{-|\}$, $A \in \Gamma$, δ contains exactly one transition of the form

$$((p, a, A), (q, B)) \text{ or } ((p, \epsilon, A), (q, B))$$

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- $\perp$ is always at the bottom of the stack

All transitions involving $\perp$ should be

$$((p, a, \perp), (q, B\perp))$$

Acceptance is by final state:

$$(s, x\textcolor{blue}{-}, \perp) \xrightarrow{*}_M (q, \epsilon, B)$$

Deterministic PDA.

$$M = (Q, \Sigma, \Gamma, \delta, \perp, \dashv, s, F)$$

- $\dashv$ is a special symbol not in Σ . Right end marker

$$\delta \subseteq (Q \times (\Sigma \cup \{\dashv\} \cup \{\epsilon\}) \times \Gamma) \times (Q \times \Gamma^*)$$

- For any $p \in Q$, $a \in \Sigma \cup \{\dashv\}$, $A \in \Gamma$, δ contains exactly one transition of the form

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All transitions involving $\perp$ should be

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Properties:

- DCFLs are closed under complementation.
- DCFLs are not closed under union, intersection.

$$\text{DCFLs} \subsetneq \text{CFL}.$$